

of 400cm² electrode area with a specific energy of 140Wh/kg that could achieve 1000 cycles.

Renewed interest has been growing in iron-air battery technology over the last few years due to tremendous advancements in nano-technology. Now it is possible to use nanostructured catalysts and higher surface area iron (Fe) nanoparticles in electrodes. With these techniques, Fe-air batteries can be made to realise a higher energy density.

How it works

The basic principle of operation of an iron-air battery is reversible rusting, as shown in Fig. 1. While discharging, the battery breathes in oxygen from the air and converts iron into solid metal oxide, i.e., rust. While charging, the application of an electrical current converts the rust back to iron, and the battery breathes out oxygen.

The energy itself is stored in the rust since that is a bond between iron and oxygen. The device's fundamental working concept is to electro-chemically reduce oxygen from the air and oxidise the metal electrode, resulting in the formation of solid metal oxides which can be recycled.

Let us look at Fig. 2 to understand the process in simple language. When discharging electricity, the battery inhales oxygen from the air and converts pure iron into rust. This allows electrons to flow, as seen on the right side of Fig. 2. As the battery is charged, the rust exhales oxygen, and oxygen exits as bubbles. Rust, i.e., oxides of iron, converts back to iron. The battery relies on this reverse rust cycle to ultimately store electricity at one-tenth the cost of a lithium-iron battery, as claimed by Form Energy, the developer of this technology.

In Fe-air batteries, Fe changes into ions on the anode, while oxygen transforms into hydroxide ions at the cathode, as shown in Fig. 3. The utilisation of oxygen from the ambi-

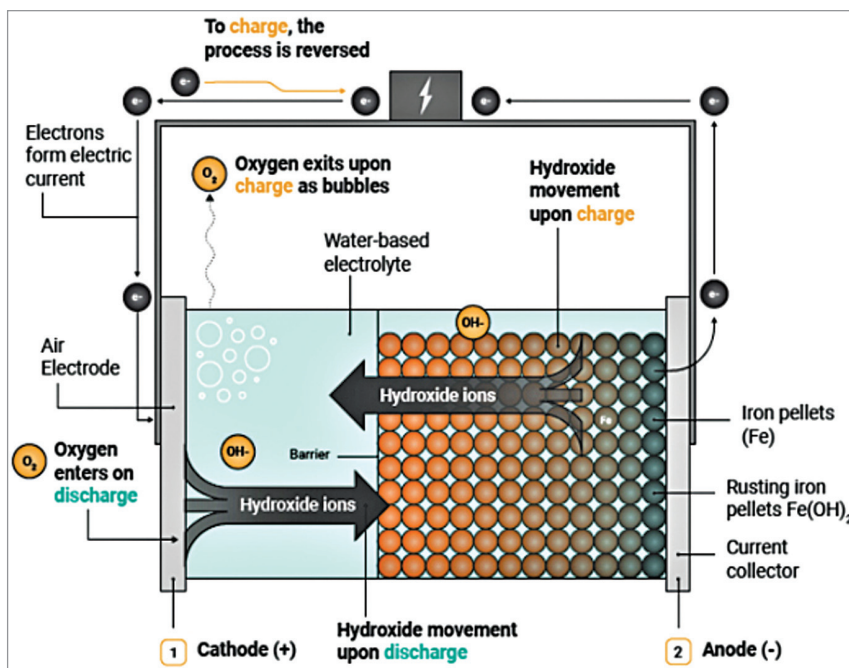


Fig. 2: Iron-air technology

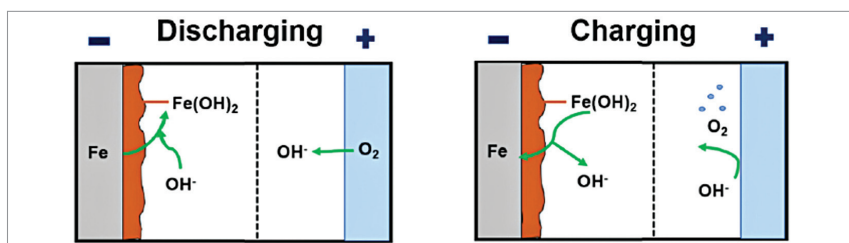


Fig. 3: Scheme of charging and discharging cycle

ent air has the benefit of lowering the cost and weight of the battery considerably. The device consists of three components, namely an anode, a cathode, and an electrolyte.

Metallic iron is used as the anode, which is the negative electrode. The chemical activity of the anode determines the discharge capacity of the anode. Because of the high activity of the metal, a side reaction with various components of the electrolyte may occur and may affect the battery efficiency.

Anodes made of iron are compatible with aqueous electrolyte, facilitating ion transfer between two electrodes, i.e., cathode and anode. An oxidation reaction, i.e., electron

loss, occurs on the anodic electrode. The released electrons, having gained excitation, become mobile and flow through a circuit to the cathodic electrode, where reduction occurs.

On the other hand, charges flow freely between the electrodes of the battery via the electrolyte. While iron anodes are compatible with aqueous electrolytes, the system requires a hydrophobic protective layer to prevent electrolyte leakage.

The cathode is the third part of the battery, and the active material for the cathode is oxygen from the air, which is freely available. It does not require any hefty shell to contain it, increasing the device's efficiency. The air cathode gener-